Loan Default Prediction System

LoanGuard AI

An ML-Based Approach to Loan Risk Assessment

TEAM MEMBERS

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ACADEMIC DETAILS

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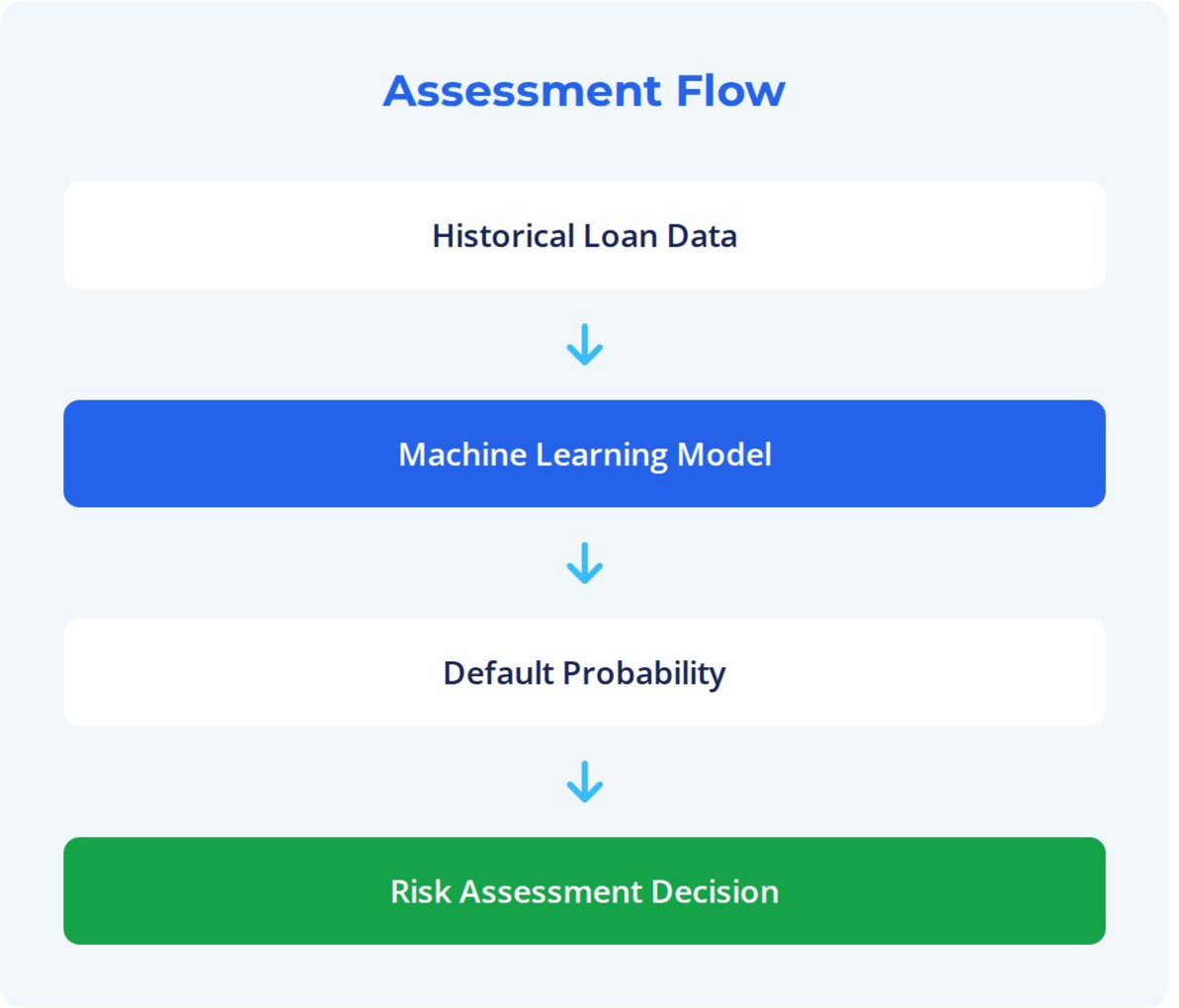
Dept: Computer Science & IT

University: K.L. University

Year: 2026–2027

INTRODUCTION

Loan assessment is a critical process for financial institutions, as applicants vary widely in their income, credit history, and repayment capability. Historical loan records hold valuable hidden patterns that reveal potential default indicators. By leveraging machine learning to analyze these patterns, **LoanGuard AI** moves beyond static rules to provide accurate probability estimation and interactive risk scoring through a web application.



PROBLEM STATEMENT

Operational Challenges

Manual evaluation is time-consuming. Large datasets defy consistent manual analysis.

Modelling Challenges

Fixed rules fail to capture complex applicant relationships and hidden patterns.

Data & Governance

Missing values, historical bias, and data leakage produce misleading predictions.

CORE PROBLEM STATEMENT

“Develop a machine learning system that estimates loan default probability from relevant applicant information and presents the prediction through an interactive web application.”

MOTIVATION

Project Drivers

- **Applied ML:** Solve a high-stakes financial problem.
- **Data Discovery:** Extract insights from historical loan records.
- **Algorithm Comparison:** Evaluate classification methodologies.
- **Full Lifecycle:** Connect a trained model to a live React web interface.
- **Auditability:** Maintain prediction history for transparent analysis.



Real-World Value

Enabling efficient, consistent, and data-driven risk estimation to protect lenders and empower analysts.

OBJECTIVES

"To develop a web-based loan default prediction system using machine learning."

1

Understand Data

Clean, preprocess, and identify targets.

2

Train Models

Train, evaluate, and select classification algorithms.

3

Predict Risk

Estimate new applicant default probabilities.

4

Explain Results

Display configurable risk recommendations.

5

Preserve History

Store predictions in Excel tracking logs.

EXISTING SYSTEM

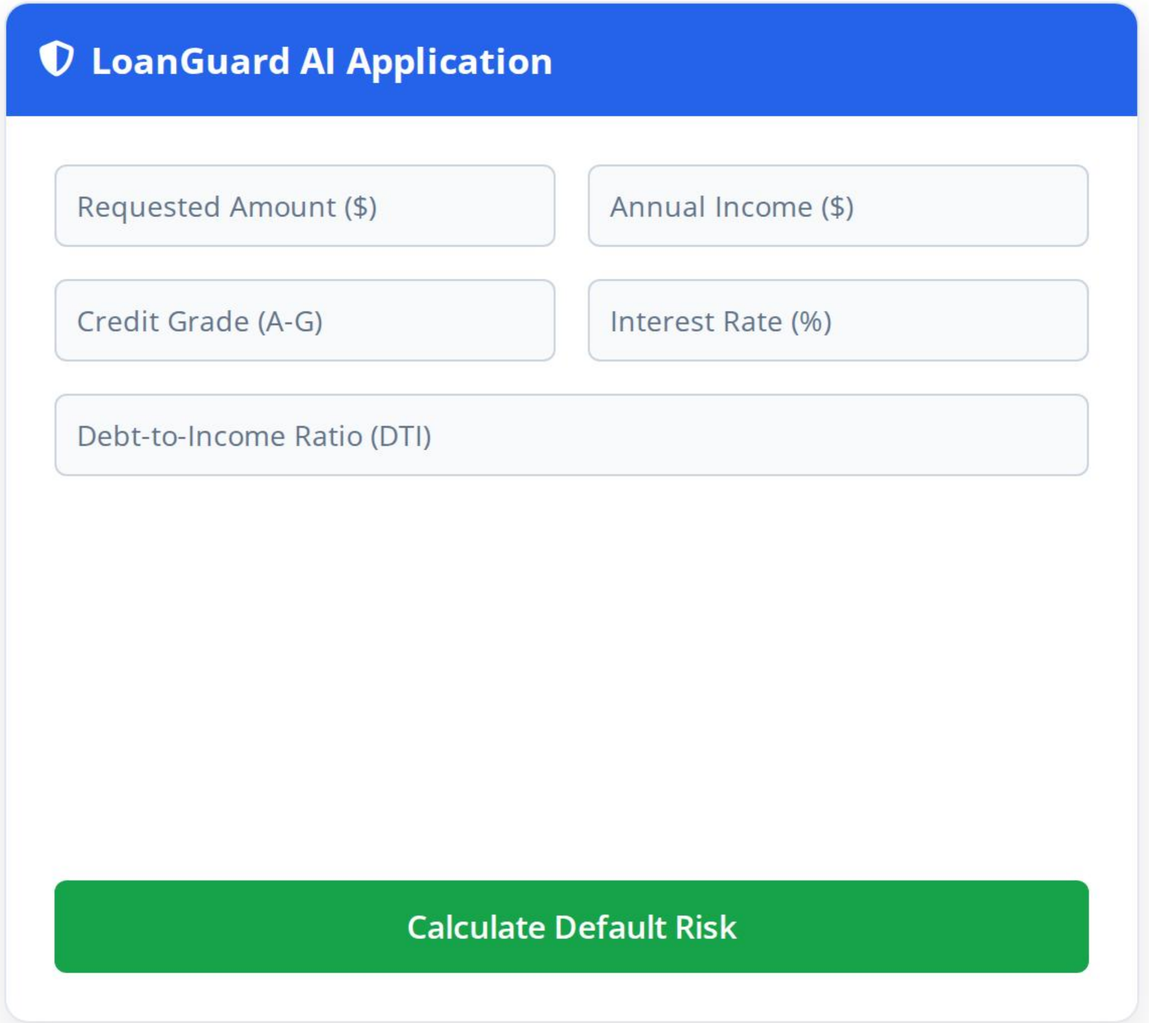
Existing Approaches	ML-Assisted Approach (Proposed)
✕ Manual document review.	✓ Automated processing via React UI.
✕ Simple credit scoring formulas.	✓ Multivariate probability estimation.
✕ Fixed rigid eligibility rules.	✓ Pattern-driven algorithmic decisions.
✕ Slow manual decision cycles.	✓ Instantaneous application feedback.
✕ Misses complex non-linear factors.	✓ Captures hidden data relationships.

Note: Existing implementations vary; automated decisions still require human validation oversight.

PROPOSED SYSTEM

Introducing LoanGuard AI

- **React Frontend:** Clean interface for applicant data entry.
- **FastAPI Backend:** High-performance request processing.
- **Saved Pipeline:** Utilizes a pre-trained ML model—no static if/else rules.
- **Probability Output:** Returns exact estimated default probability.
- **Configurable Logic:** Maps probability to adjustable risk categories.
- **Excel Storage:** Securely logs completed prediction histories.



SYSTEM ARCHITECTURE / BLOCK DIAGRAM

Training Workflow

loan_final313.csv

↓

Data Inspection & Cleaning

↓

Feature Selection & Leakage Prevention

↓

Preprocessing & Split

↓

Model Training & Evaluation

↓

Saved ML Pipeline: loan_default_model.joblib

Prediction Workflow (Inference)

Applicant Data Input

↓

React Frontend ↔ FastAPI Backend

↓

Loads Saved ML Model (No retraining)

↓

Probability & Prediction Generation

↓

Risk & Recommendation Logic (JSON Response)

↓

Excel History: prediction_history.xlsx

METHODOLOGY / WORKFLOW

01. Load Dataset

02. Explore Columns & Types

03. Identify Target Variable

04. Clean Data & Missing Vals

05. Encode Categoricals

06. Scale Numerical Features

07. Prevent Data Leakage

08. Train/Test Split

09. Train Classifiers

10. Evaluate & Compare

11. Select & Save Model

12. Develop FastAPI Backend

13. Develop React Frontend

14. Connect APIs

15. Test Predictions

16. Finalize Excel Logging

Crucial Step: Preprocessing transforms are strictly fitted on training data and reused consistently during prediction.

TECHNOLOGY STACK

Frontend

React, Vite, HTML5, CSS3, JavaScript.

Backend

Python, FastAPI, Uvicorn.

Machine Learning

scikit-learn, XGBoost.

Data Processing & Storage

Pandas, NumPy, Joblib, OpenPyXL, CSV, Excel.

Development Tools

VS Code, Git, GitHub, Matplotlib.

SYSTEM MODULES / COMPONENTS

1. User Interface

Collects applicant details.

2. Data Collection

Loads historical CSV records.

3. Preprocessing

Encodes and handles missing values.

4. ML Training

Trains classification models.

5. Prediction

Executes saved pipeline inference.

6. Decision

Maps probability to risk logic.

7. Excel Storage

Appends successful predictions.

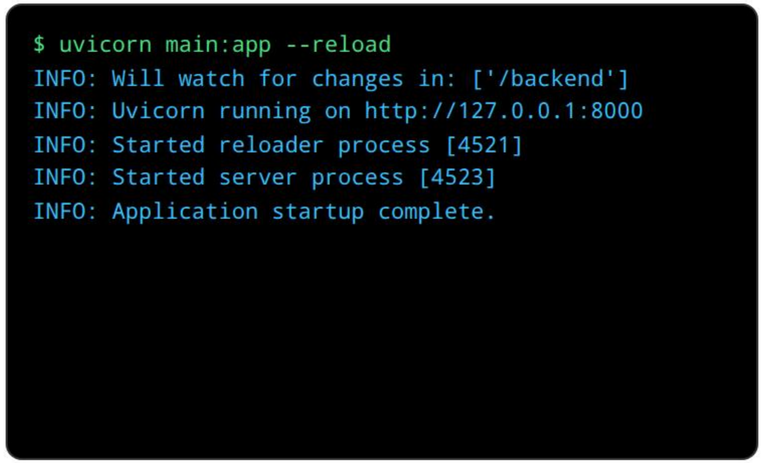
8. Clustering

Demonstrates K-Means grouping.

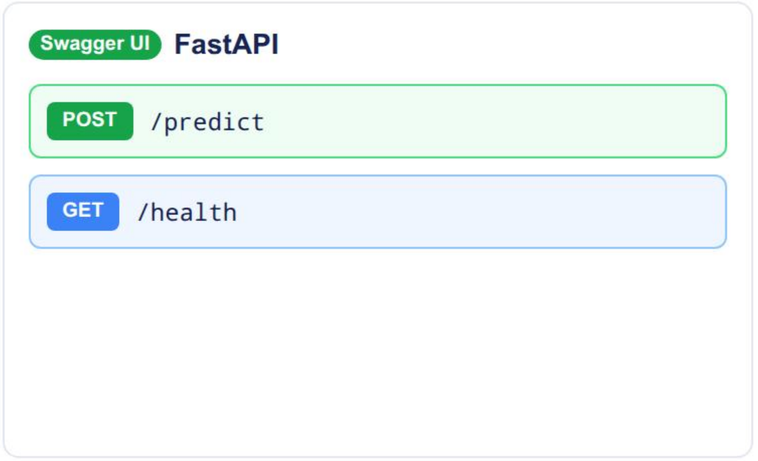
IMPLEMENTATION



1. Project Folder Structure



2. Backend Terminal



3. FastAPI Documentation



4. React UI Running



5. Final Prediction Result



ID	Amount	Prob	Risk
101	$15K	12.4%	Low
102	$8.5K	36.8%	Med
103	$24K	78.1%	High

6. Excel Storage Log

DATASET / INPUT DATA

Dataset Statistics

- **Name:** loan_final313.csv
- **Records:** ~887,379 rows
- **Columns:** ~30 features
- **File Size:** ~151.10 MB

Data Leakage Prevention

Excluded post-loan fields (e.g., total_pymnt, recoveries, final_d) that improperly reveal the outcome.

Preprocessing & Features

- **Target Variable:** Verified label mapping applied.
- **Cleaning:** Handled missing values systematically.
- **Encoding:** Categorical features processed cleanly.
- **UI Inputs:** Requested amount, Income, Grade, Interest Rate, DTI ratio.

ALGORITHM / MODEL

1. Logistic Regression

Baseline binary classification using sigmoid probability.

2. Decision Tree

Feature-based splits. Easy to interpret, risks overfitting.

3. Random Forest

Robust ensemble method using bagging and tree combinations.

4. XGBoost

Gradient boosting. Sequential tree building with regularization.

Unsupervised Extension (If Implemented)

K-Means: Exploratory clustering. Evaluated via Silhouette score and Elbow method. Does not replace the core default prediction.

RESULTS AND EVALUATION

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	[Insert %]	[Insert %]	[Insert %]	[Insert %]	[Insert %]
Decision Tree	[Insert %]	[Insert %]	[Insert %]	[Insert %]	[Insert %]
Random Forest	[Insert %]	[Insert %]	[Insert %]	[Insert %]	[Insert %]
XGBoost (Selected)	**[Insert %]**	**[Insert %]**	**[Insert %]**	**[Insert %]**	**[Insert %]**

Demonstration Result (Single Prediction Example)

Inference Model: XGBoost | **Default Probability:** 36.8%

Risk Level: Medium Risk | **Decision:** Eligible for Loan (Conditional Review)

SCREENSHOTS / DEMO

A. Applicant Form

Enter Applicant Details

Amount: 15000

Income: 65000

Grade: B

Interest: 10.5%

Predict Risk

B. Prediction Result

Default Probability

36.8%

Medium Risk

C. Excel History

prediction_history.xlsx

Timestamp	Model	Prob	Decision
2026-10-13 14:02	XGBoost	12.4%	Approve
2026-10-13 14:15	XGBoost	36.8%	Review
2026-10-13 14:28	XGBoost	78.1%	Reject

D. FastAPI Endpoints

LoanGuard API 1.0.0

POST /predict

GET /health

ADVANTAGES

Instant Prediction

Automated assessment immediately after submission.

Probability Transparency

Outputs exact default probability, not just binary flags.

User-Friendly Web UI

Clean, accessible React frontend for simple operation.

Built-in Auditability

Automatically records every completed prediction in Excel.

Decoupled Architecture

Strict separation between training and production prediction.

End-to-End Educational ML

Demonstrates complete lifecycle from CSV to deployed app.

APPLICATIONS

Potential Use Cases

- **Initial Screening:** First-pass filtering for loan applicants.
- **Financial Analytics:** Pattern recognition in portfolios.
- **Historical Analysis:** Validating past lending decisions.
- **ML Research:** Baseline for future algorithmic tuning.
- **FinTech Prototyping:** Blueprint for decision support tools.

Target Users

Students

Researchers

Data Analysts

FinTech Developers

Institutions Exploring ML

Disclaimer: This project is an educational prototype, not an officially approved banking system.

LIMITATIONS

1. Data Quality

Prediction accuracy heavily dependent on historical data quality. Past data may not reflect future applicants.

2. Fairness & Bias

Historical datasets may contain ingrained demographic bias. Further fairness testing is required.

3. Prediction Constraints

No absolute guarantee of repayment. Thresholds are configurable project settings, not rigid industry standards.

4. Storage & Scaling

Excel logging is suitable for demos but lacks DB scale. Dataset (>151MB) exceeds GitHub standard limits.

FUTURE SCOPE AND CONCLUSION

Future Scope

- → Incorporate diverse datasets for robustness.
- → Add SHAP-based model explanations.
- → Implement secure user authentication.
- → Migrate from Excel to SQL DB storage.
- → Generate downloadable analytics reports.
- → Perform deep fairness and privacy testing.

Conclusion

LoanGuard AI successfully demonstrates applied machine learning for default probability estimation.

It unifies historical preprocessing, saved pipeline inference, a responsive React frontend, and a FastAPI backend.

By outputting configurable risk recommendations and preserving prediction history, it serves as a complete educational end-to-end ML project.

LoanGuard AI — Turning Loan Data into Risk Insights.

THANK YOU | Questions & Answers